

Given the following measures on the Statistics scores for a group of 25 second year's students:

The smallest 3 values	The first quartile	The median	The third quartile	The largest 3 values
30, 33, 37	40	45	60	72, 80, 98

- 1) Draw the Box-Plot.**
- 2) Find a measure of central tendency**
- 3) Is the value "15" an outlier? Why?**
- 4) Determine the direction of the skewness (without calculation).**
- 5) Calculate the coefficient of skewness, and comment.**
- 6) Is it a difficult exam? Why?**

Some Exercises

- (1) What name is given to a table that lists all the values that a discrete random variable X can assume and their corresponding probabilities? Probability Mass Function (PMF)
- (2) For the probability distribution of a discrete random variable, the probability of any single value of X is always
 - a. In the range 0 to 1 b. 1.0 c. less than zero
- (3) For the probability distribution of a discrete random variable, the sum of the probabilities is always 1.....
- (4) The parameters of the binomial probability distribution are ... n and ... p
- (5) The binomial distribution is skewed to the right if $\pi \leq 0.5$
- (6) The parameter/ parameters of the Poisson probability distribution is/ are ... λ (num of occurrence).....

- (7) Find the mean and the standard deviation from the followin table Mean $= (-4)(0.2) + 0(0.3) + 1(0.3) + 2(0.2) = -0.1$

x	-4	0	1	2
$P(X = x)$	0.2	0.3	0.3	0.2

$$E(X^2) = \sum x^2 P(X=x) = (-4)^2(0.2) + (0)^2(0.3) + (1)^2(0.3) + (2)^2(0.2) = 4.3$$

$$\text{Variance} = 4.3 - (-0.1)^2 = 4.29$$

$$\text{standard dev} = \sqrt{4.29} = 2.071$$

- (8) A factory has eight machines. The probability is 0.04 that any machine will break down at any time. Find the probability that at any given time:
 - a. All eight machines will be broken down $C(8,8) * (0.04)^8 * (0.96)^0 =$
 - b. Exactly two machines will be broken down $C(8,2) * (0.04)^2 * (0.96)^6 = 0.2007$
 - c. None of the machines will be broken down $C(8,0) * (0.04)^0 * (0.96)^8 = 0.6634$
- (10) A high school boys' basketball team averages 1.2 technical fouls per game. Find the probability that in agiven game this team will commit: $\lambda = 1.2$
 - a. Exactly three technical fouls $e^{-1.2} * (1.2)^3 / 3! = 0.0862$
 - b. At least two technical fouls $P(X \geq 2) = 1 - P(X=0) - P(X=1) = 1 - e^{-1.2} (1 + 1.2) = 0.337$
 - c. Find the mean and the standard deviation Mean $= \lambda = 1.2$ Standard Deviation $= \sqrt{\lambda} = \sqrt{1.2} = 1.095$